
IAL Biology Unit 1

WBI11

Molecules, Diet, Transport and Health

A* revision and exam-technique guide

Built from analysis of 22 past papers and mark schemes,
January 2019 to May 2026

WBI11/01 · 80 marks · 1 hour 30 minutes
40% of the IAS · 20% of the full IAL

1 • What the papers actually look like

Across all 22 papers the structure is near-identical: **eight questions, 80 marks, no separate multiple-choice section**. One paper in 2023 and one in 2026 had nine. Multiple-choice items appear only occasionally, embedded inside questions — typically 0–3 per paper. This is a written-answer paper, not a recall quiz.

Marks ramp steeply through the paper

Question	Mean marks	Range observed	Implication
Q1	5.7	5–8	Short, usually single-topic. Finish these fast — they are the cheapest marks on the paper and should take well under a minute per mark.
Q2	6.8	5–10	
Q3	8.2	5–10	
Q4	8.7	6–11	
Q5	10.7	8–13	Multi-part, data-heavy, often spanning two or three spec areas plus a calculation and an extended-response part. Budget the bulk of your time here.
Q6	11.6	9–15	
Q7	13.0	10–15	
Q8	14.1	11–15	

TIMING
80 marks in 90 minutes. Q1–Q4 average 29 marks; aim to clear them in **28 minutes**, leaving **50 minutes** for Q5–Q8 and **12 minutes** to check. Most candidates who run out of time do so because they over-wrote on the early short questions.

Command words, counted across all 22 papers

Command	Per paper	What it requires
Explain	9.6	By far the dominant command. Requires a reason , not a description. If your sentence contains no "because", "so", "therefore" or "due to", it is probably not an explanation.
Calculate	3.7	Guaranteed every paper, usually 3–5 separate calculations. Show working; most carry a method mark.
Describe	2.9	Say what happens or what the data shows. No reason needed — adding one wastes time but costs nothing. Quote figures from graphs and tables with units.
Suggest	2.8	You are not expected to have learned this. Apply principles to an unfamiliar context. Never leave these blank — a reasoned attempt frequently scores.
Give / Name / State	5.0 combined	One-word or one-phrase answers. Do not explain.
Complete / Draw	3.0	Tables, genetic diagrams, graph axes, membrane or molecule diagrams.
Compare	0.8	Must address both items in every sentence — "A has X whereas B has Y". A sentence about only one scores nothing.
Discuss / Comment	1.2	Usually the extended-response part. Needs both sides and a judgement.

The practical consequence: roughly **half the marks on this paper are explanation marks**. Content knowledge alone is not enough — the gap between B and A* is almost entirely in how precisely explanations are written.

Topic frequency across the 22 papers

Counted by how many papers gave each area substantial coverage:

Topic area	Papers (of 22)	Verdict
Maths and data-handling skills	22	Guaranteed. Every paper, usually several marks each. These are the areas where preparation pays most reliably.
Carbohydrates	20	
Blood vessels and circulation	18	
Diet, energy, cholesterol and lipoproteins	18	
Proteins and amino acids	17	
Enzymes	16	Very likely. Appears in roughly two papers in three. Expect at least three of these six on any paper.
Atherosclerosis and blood clotting	16	
Lipids	15	
Cell membranes and transport	14	
Protein synthesis	13	
Mutation; inheritance and genetic diagrams	12	Common. Half of all papers.
CVD risk factors and epidemiology	11	
CVD treatments and side effects	11	
DNA structure and replication	11	
Heart and cardiac cycle	8	
Cystic fibrosis and CFTR	7	Periodic — but when they appear they carry heavy marks, especially the cardiac cycle with its timing calculations.
Gas exchange and lungs	6	
Water as a biological molecule	2	
Genetic screening and ethics	2	
Gene therapy	1	

WHAT THIS TABLE MEANS FOR REVISION

Carbohydrates, circulation, lipids and proteins between them account for a large share of every paper, and they are all pure recall plus structured explanation. Prioritise them. But do not use the bottom of the table as permission to skip — the whole specification is examinable, and gene therapy appearing once in 22 papers means it will appear again.

2.1 Water

Water is a **dipolar** molecule: oxygen is more electronegative than hydrogen, so the oxygen carries a partial negative charge and the hydrogens partial positive. This allows **hydrogen bonds** between water molecules, which explains every property below.

Property	Cause	Biological importance
Excellent solvent	Polar; attracts and surrounds charged and polar solutes	Transport medium in blood and cytoplasm; metabolic reactions occur in solution
High specific heat capacity	Much energy needed to break hydrogen bonds	Buffers temperature change in cells and in organisms with high water content
High latent heat of vaporisation	Energy needed to break hydrogen bonds on evaporation	Sweating gives efficient cooling for little water loss
Cohesion and high surface tension	Hydrogen bonds between molecules	Continuous water columns in transport; surface habitat for small organisms
Ice less dense than water	Hydrogen bonds hold molecules in an open lattice	Ice insulates water beneath; aquatic organisms survive

2.2 Carbohydrates

Present in 20 of 22 papers — the single most reliably examined molecule topic.

- **Monosaccharides** — glucose, fructose, galactose. General formula (CH₂O)_n. Glucose exists as **α-glucose** (OH below C1) and **β-glucose** (OH above C1).
- **Condensation** joins two monosaccharides, forming a **glycosidic bond** and releasing **one molecule of water**. **Hydrolysis** is the reverse and **uses** a water molecule.
- **Disaccharides** — maltose (glucose + glucose), sucrose (glucose + fructose), lactose (glucose + galactose).

Polysaccharide	Structure	Why the structure suits the function
Amylose	α-glucose joined by 1,4 glycosidic bonds only; unbranched; coils into a helix	Compact, so a large amount of glucose is stored in a small space; insoluble
Amylopectin	α-glucose; 1,4 and 1,6 glycosidic bonds; branched	Branching gives many ends for enzymes to act on, so glucose is released rapidly
Glycogen	α-glucose; 1,4 and 1,6 bonds; more highly branched than amylopectin	Even faster hydrolysis, suiting the higher metabolic rate of animals; stored in liver and muscle

THE STANDARD 4-MARK ANSWER: WHY POLYSACCHARIDES ARE GOOD STORAGE MOLECULES

They are **insoluble**, so they have **no osmotic effect** on the cell and water is not drawn in. They are **compact**, so a large quantity is stored in a small volume. They are **branched**, giving many ends for rapid hydrolysis when glucose is needed. They are **large**, so they cannot diffuse out of the cell.

2.3 Lipids

- **Triglyceride** = one **glycerol** + three **fatty acids**, joined by **ester bonds** formed in condensation reactions releasing **three** water molecules.
- **Saturated** fatty acids have **no C=C double bonds**; the chains are straight, pack closely, and have stronger intermolecular attractions, so they are **solid at room temperature**.
- **Unsaturated** fatty acids have **one or more C=C double bonds** causing **kinks** in the chain; they pack less closely, so they are **liquid at room temperature** (oils). Monounsaturated has one double bond; polyunsaturated has more than one.
- Lipids store **more energy per gram** than carbohydrates because they contain **proportionally more C-H bonds and less oxygen** — they are more reduced.
- **Phospholipids** have a hydrophilic phosphate head and two hydrophobic fatty acid tails, which is why they form bilayers (\$3.2).

2.4 The heart and cardiac cycle

Appears in only 8 of 22 papers, but carries heavy marks when it does, and almost always with a timing calculation attached.

- The **left ventricle has a thicker muscular wall** than the right because it must generate higher pressure to pump blood around the **whole body**, whereas the right pumps only to the lungs — at lower pressure, which also prevents damage to the delicate alveolar capillaries.
- **Atrioventricular valves** (tricuspid right, bicuspid/mitral left) prevent backflow into the atria. **Semilunar valves** (pulmonary and aortic) prevent backflow into the ventricles.
- **Valves open and close because of pressure differences** — a valve opens when the pressure behind it exceeds the pressure in front. That sentence is the mark on almost every valve question.
- The **coronary arteries** supply the heart muscle itself with oxygenated blood; their blockage causes myocardial infarction.

Stage	What happens	Valves
Atrial systole	Atria contract, pushing remaining blood into the ventricles	AV valves open; semilunar closed
Ventricular systole	Ventricles contract; pressure rises sharply above that in the atria, then above that in the arteries	AV valves close (first heart sound), then semilunar valves open
Diastole	All chambers relax; pressure falls; blood flows passively into the atria and on into the ventricles	Semilunar valves close (second heart sound); AV valves open

Cardiac output = stroke volume × heart rate. Timing questions typically give a pressure–time graph and ask for the duration of one cycle, the proportion of the cycle spent in ventricular systole, or the heart rate in beats per minute (60 ÷ cycle length in seconds).

2.5 Blood vessels, circulation and tissue fluid

In 18 of 22 papers. Learn the structure–function links as matched pairs.

Vessel	Structure	Function it serves
Artery	Thick wall with much elastic tissue and smooth muscle ; narrow lumen; no valves; endothelium folded	Withstands and maintains high pressure ; elastic tissue stretches and recoils to smooth the pulsatile flow between heartbeats
Arteriole	More smooth muscle, less elastic tissue	Vasoconstriction and vasodilation redirect blood flow to where it is needed
Capillary	Wall one cell thick (squamous endothelium); very narrow lumen; pores between cells; huge total cross-sectional area	Short diffusion distance and large surface area for exchange; the large total cross-sectional area slows the blood , allowing time for exchange

Vein	Thin wall, little elastic tissue or muscle; wide lumen ; valves	Low pressure; wide lumen reduces resistance; valves prevent backflow , and contraction of surrounding skeletal muscle squeezes blood onwards
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TISSUE FLUID FORMATION — A RECURRING EXPLANATION

At the **arterial end** of a capillary, the **hydrostatic pressure** of the blood exceeds the opposing osmotic effect of the plasma proteins, so fluid is **forced out** through the capillary wall. Plasma proteins and cells are too large to leave.

At the **venous end**, hydrostatic pressure has fallen, while the plasma proteins left behind have **lowered the water potential** of the blood. Water therefore **returns by osmosis**. Excess tissue fluid drains into the **lymphatic system**.

2.6 Atherosclerosis and blood clotting

THE ATHEROSCLEROSIS SEQUENCE

1. The **endothelium is damaged** — by high blood pressure, or by toxins in cigarette smoke.
2. An **inflammatory response** follows; white blood cells move into the artery wall.
3. These accumulate **cholesterol and lipids**, forming a **fatty streak**.
4. Over time this builds into an **atheroma or plaque**, with fibrous tissue and calcium salts hardening it.
5. The plaque **narrows the lumen**, so **blood pressure rises** — which causes further endothelial damage. This is a **positive feedback** cycle, and saying so earns the mark.

Consequences: angina (narrowed coronary arteries, reduced oxygen to heart muscle); myocardial infarction (complete blockage, cardiac muscle dies); stroke (blockage or rupture in an artery supplying the brain); aneurysm (weakened artery wall balloons and may rupture).

THE CLOTTING CASCADE — LEARN IT IN ORDER

THREE STEPS, THREE MARKS

1. Damage to the vessel causes platelets and damaged tissue to release **thromboplastin**.
2. Thromboplastin, in the presence of **calcium ions** and vitamin K, converts **prothrombin into thrombin**.
3. Thrombin acts as an enzyme catalysing the conversion of **soluble fibrinogen into insoluble fibrin**, which forms a **mesh** trapping platelets and red blood cells to form the clot.

A clot forming inside an intact vessel is a **thrombus**; if it breaks free and lodges elsewhere it is an **embolus**.

2.7 Risk, correlation and epidemiology

This is the area where candidates most often write confidently and score nothing, because the required answer is sceptical rather than enthusiastic.

THE ANSWER TO "DOES THIS DATA SHOW THAT X CAUSES Y?"

It shows a **correlation**, but correlation **does not prove causation**. Then give at least two of the following, chosen to fit the data given:

- Other **confounding variables** may be responsible, and were not controlled.
- The **sample size** may be too small, or the **sample unrepresentative** of the population.
- There is no **control group**.
- The study shows an **association over a population**, not a mechanism in an individual.
- No indication of whether the difference is **statistically significant**.
- The data are from a single time period, place or age group.

CVD risk factors. Non-modifiable: age, sex (male higher until female menopause), genetics and family history. Modifiable: smoking, high blood pressure, high blood cholesterol, obesity, physical inactivity, diet high in saturated fat and salt, excessive alcohol, diabetes.

2.8 Diet, cholesterol and energy balance

	LDL — low-density lipoprotein	HDL — high-density lipoprotein
Carries	Cholesterol from the liver to the tissues	Cholesterol from the tissues to the liver for breakdown and excretion
Mechanism	Binds to LDL receptors on cell membranes. A diet high in saturated fat saturates these receptors , so excess LDL remains circulating in the blood and is deposited in artery walls.	Reduces blood cholesterol concentration and can help remove cholesterol from plaques
Effect on CVD risk	Increases risk	Reduces risk

Associated measures: **BMI = mass (kg) ÷ height (m)²**; waist-to-hip ratio as an indicator of central obesity; energy budget as energy intake versus energy expenditure, with a sustained surplus stored as fat. **Antioxidants** such as vitamin C are thought to reduce damage caused by free radicals.

2.9 Treating cardiovascular disease

Drug class	How it works	Risks and side effects
Antihypertensives beta blockers, ACE inhibitors, diuretics, calcium channel blockers	Lower blood pressure, reducing endothelial damage and the workload on the heart	Fatigue, dizziness, headaches, swollen ankles; several drugs often needed together
Statins	Inhibit the enzyme that catalyses cholesterol synthesis in the liver, reducing blood LDL	Muscle pain and weakness, liver damage, digestive problems
Anticoagulants warfarin, heparin	Reduce formation of clots in the blood vessels	Uncontrolled bleeding after injury; dosage must be monitored carefully
Platelet inhibitors aspirin, clopidogrel	Reduce the tendency of platelets to stick together	Excessive bleeding, stomach irritation and ulcers

Questions often ask you to **evaluate** whether a treatment should be prescribed. Balance the reduction in risk against the side effects, note that lifestyle change carries no side effects, and state a judgement — an answer with no judgement does not reach the top band.

3 · Topic 2 — Genes and Health

3.1 Gas exchange and Fick's law

FICK'S LAW — QUOTE IT, THEN APPLY IT

Rate of diffusion ∝ (surface area × concentration difference) ÷ diffusion distance

Every adaptation of an exchange surface maps onto one of the three terms:

- **Large surface area** — many alveoli, each folded; total area roughly the size of a tennis court.
- **Short diffusion distance** — the alveolar wall and the capillary endothelium are each **one cell thick** (squamous epithelium), so the barrier is two cells.
- **Steep concentration gradient maintained** — by **ventilation** bringing in fresh air and by **continuous blood flow** carrying oxygen away.

As an organism gets larger its **surface area to volume ratio falls**, so diffusion across the body surface becomes too slow to meet demand — which is why a specialised exchange surface and a mass transport system become necessary. *SA:V* calculations appear regularly.

3.2 Cell membranes and transport

The **fluid mosaic model**: a **phospholipid bilayer** with hydrophilic phosphate heads facing the aqueous surroundings and hydrophobic fatty acid tails facing inwards. "Fluid" because the phospholipids move laterally; "mosaic" because proteins are scattered through it.

- **Channel and carrier proteins** — transport of polar and charged substances.
- **Cholesterol** — sits between phospholipids, regulating fluidity and stability.
- **Glycoproteins and glycolipids** — cell recognition, receptors, cell adhesion.

Process	Energy	Direction	Mechanism and examples
Simple diffusion	Passive	Down the concentration gradient	Small or non-polar molecules pass directly through the bilayer — oxygen, carbon dioxide
Facilitated diffusion	Passive	Down the concentration gradient	Through channel or carrier proteins ; for polar or charged species — ions, glucose, amino acids
Osmosis	Passive	From higher to lower water potential	Movement of water across a partially permeable membrane . Pure water has a water potential of zero ; adding solute makes it more negative .
Active transport	Requires ATP	Against the concentration gradient	Carrier protein changes shape using energy from ATP hydrolysis
Endo- and exocytosis	Requires ATP	Bulk, either direction	Vesicles fuse with or pinch off from the membrane; for large molecules and particles

DEFINITIONS WORTH BEING EXACT ABOUT

Osmosis — the movement of water molecules from a region of **higher water potential** to a region of **lower water potential** through a **partially permeable membrane**.

Active transport — the movement of substances **against** a concentration gradient, using **carrier proteins** and energy from **ATP**.

3.3 Cystic fibrosis

CF is the specification's worked case study, linking a single gene to a protein to a symptom. Questions nearly always require the full chain.

THE CHAIN, FROM GENE TO SYMPTOM

The **CFTR gene** on chromosome 7 codes for the **CFTR protein**, a **chloride ion channel** in the cell surface membrane of epithelial cells. The commonest mutation is a **deletion of three bases**, so the protein is missing one amino acid (phenylalanine) and folds incorrectly.

Faulty or absent CFTR means **chloride ions are not secreted** out of the cell. **Sodium ions** are then absorbed from the mucus into the cell, so **water follows by osmosis** out of the mucus. The mucus therefore becomes **thick and sticky**.

System	Effect of the thick mucus	Consequence
Gas exchange	Mucus blocks bronchioles and is hard to clear by cilia; it accumulates and traps bacteria	Reduced surface area and increased diffusion distance, so less gas exchange; repeated infections; alveolar damage
Digestive	Mucus blocks the pancreatic duct , so digestive enzymes cannot reach the small intestine; cysts form in the pancreas	Food not fully digested or absorbed, so poor growth and weight loss; enzyme supplements are given with meals
Reproductive	In males the sperm duct is blocked or absent; in females the cervical mucus thickens	Reduced fertility in both sexes

3.4 Proteins and amino acids

An amino acid has a central carbon bonded to an **amine group (-NH₂)**, a **carboxyl group (-COOH)**, a hydrogen, and a variable **R group**. Amino acids join by **condensation**, forming a **peptide bond** and releasing water.

Level	What it is	Bonds involved
Primary	The sequence of amino acids in the polypeptide	Peptide bonds
Secondary	α-helix or β-pleated sheet	Hydrogen bonds between the C=O and N-H groups of the backbone
Tertiary	The overall three-dimensional folding of one polypeptide	Hydrogen bonds, ionic bonds , disulfide bridges , hydrophobic interactions — all between R groups
Quaternary	More than one polypeptide chain, sometimes with a prosthetic group	The same bond types, between chains

	Globular	Fibrous
Shape	Compact, roughly spherical, folded	Long, parallel chains
Solubility	Soluble — hydrophilic R groups face outwards	Insoluble
Role	Metabolic — enzymes, haemoglobin, antibodies	Structural — collagen, keratin

Collagen is the standard fibrous example: **three polypeptide chains** wound into a **triple helix**, held by hydrogen bonds, with **glycine** as every third amino acid because it is small enough to fit in the centre. Molecules are **cross-linked** and arranged with **staggered ends**, giving great tensile strength.

3.5 Enzymes

Enzymes are **globular proteins** acting as biological catalysts. They **lower the activation energy** of a reaction, so more substrate molecules have enough energy to react at a given temperature.

- The **active site** has a shape **complementary** to the substrate; they form an **enzyme-substrate complex**.
- **Lock and key** — the active site is already an exact fit. **Induced fit** — the active site **changes shape slightly** as the substrate binds, which puts strain on the substrate's bonds and lowers the activation energy further. Induced fit is the accepted model.

Factor	Effect and explanation
Temperature	Rate increases as molecules gain kinetic energy , so there are more frequent successful collisions and more enzyme–substrate complexes form. Above the optimum, hydrogen and ionic bonds holding the tertiary structure break , the active site changes shape and is no longer complementary to the substrate: the enzyme is denatured .
pH	Away from the optimum, ionic and hydrogen bonds in the tertiary structure are disrupted as R groups change charge, altering the active site shape. Extreme pH denatures.
Substrate concentration	Rate increases because more complexes form, until all active sites are occupied — the enzyme is saturated and rate plateaus at $V_{\max}$. Enzyme concentration is then the limiting factor.
Competitive inhibitor	Similar shape to the substrate; binds to the active site , blocking it. The effect is reduced by increasing substrate concentration .
Non-competitive inhibitor	Binds elsewhere on the enzyme, changing the shape of the active site so the substrate no longer fits. Cannot be overcome by increasing substrate concentration.

SAY "DENATURED", NEVER "KILLED"

Enzymes are molecules, not organisms. "The enzyme was killed" is marked wrong every time. Also note that denaturation involves **breaking the bonds that maintain the tertiary structure**, not breaking peptide bonds — the primary structure is unchanged.

3.6 DNA structure and replication

- A **nucleotide** is a **deoxyribose sugar**, a **phosphate group** and a **nitrogenous base**.
- Nucleotides join by **phosphodiester bonds** to form a sugar–phosphate backbone; two **antiparallel** strands coil into a **double helix**.
- **Complementary base pairing: A-T with two hydrogen bonds, C-G with three**. A purine always pairs with a pyrimidine, keeping the helix a constant width.

SEMI-CONSERVATIVE REPLICATION

DNA helicase breaks the **hydrogen bonds** between bases and unwinds the helix. Each strand acts as a **template**. Free nucleotides align by **complementary base pairing**, and **DNA polymerase** joins them by forming **phosphodiester bonds**. Each new molecule contains **one original strand and one new strand** — hence "semi-conservative".

Meselson and Stahl grew bacteria in heavy ^{15}N , transferred them to ^{14}N and used **density gradient centrifugation**. After one generation all the DNA was of **intermediate density**, which rules out conservative replication. After two generations there were **equal amounts of intermediate and light** DNA, which rules out dispersive replication and supports the semi-conservative model.

3.7 Protein synthesis

Stage	Location	What happens
Transcription	Nucleus	Helicase unwinds the DNA and breaks hydrogen bonds. RNA polymerase moves along the template (antisense) strand , joining complementary RNA nucleotides — with uracil in place of thymine . The pre-mRNA has its introns removed by splicing, and the mature mRNA leaves through a nuclear pore .
Translation	Ribosome	mRNA binds to a ribosome. A tRNA with an anticodon complementary to the mRNA codon brings its specific amino acid. Adjacent amino acids are joined by peptide bonds ; the ribosome moves along codon by codon until a stop codon is reached.

The **genetic code** is a **triplet** code, **non-overlapping**, **degenerate** (most amino acids are coded for by more than one codon) and near-**universal**. Degeneracy matters because it explains why many substitution mutations have no effect.

3.8 Mutation

A **mutation** is a change in the base sequence of DNA.

- **Substitution** — one base replaced. Often has **no effect**, because the code is **degenerate** and the new codon may specify the same amino acid. If it does change the amino acid, the **primary structure** changes, which may alter the **tertiary structure** and so the protein's function.
- **Deletion or insertion** — causes a **frameshift**: every subsequent codon is read differently, so most of the amino acid sequence after the mutation is wrong. Usually far more damaging than a substitution.

3.9 Inheritance and genetic diagrams

Key terms: **allele** (a version of a gene); **dominant** (expressed in the heterozygote); **recessive** (expressed only in the homozygote); **genotype** (alleles present); **phenotype** (observable characteristics); **carrier** (heterozygote for a recessive condition, unaffected but able to pass it on).

DRAWING A GENETIC DIAGRAM FOR FULL MARKS

1. Define your symbols — "F = normal allele, f = cystic fibrosis allele".
2. Write the **parental genotypes** and their phenotypes.
3. Show the **gametes**, circled.
4. Use a Punnett square for the offspring genotypes.
5. State the **phenotypes and the ratio or probability**, answering the actual question. Two carriers give a 1 in 4 (25%) probability of an affected child, and a 2 in 3 probability that an unaffected child is a carrier — the second is frequently asked and frequently got wrong.

Pedigree analysis. If two unaffected parents have an affected child, the condition is **recessive**. If the condition appears in both sexes in roughly equal numbers it is **autosomal** rather than sex-linked. Papers often then ask you to **discuss the extent to which** the pedigree proves this — the answer acknowledges the evidence but notes the **small sample** and that other patterns cannot be fully excluded.

3.10 Genetic screening, ethics and gene therapy

Technique	Method and timing	Evaluation
Amniocentesis	Needle into the amniotic sac at about 15–17 weeks; fetal cells in the amniotic fluid are cultured and tested	Reliable, but performed late, and carries a small risk of miscarriage
Chorionic villus sampling	Sample of placental tissue taken at about 8–12 weeks	Earlier result allowing an earlier decision, but a somewhat higher miscarriage risk

Preimplantation genetic diagnosis	Embryos created by IVF; a single cell is removed and tested; only unaffected embryos are implanted	Avoids a decision about termination, but IVF is expensive, invasive and uncertain, and it raises concerns about embryo selection
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Ethical considerations to deploy in a discussion question: informed consent and the right not to know; confidentiality, and possible discrimination by insurers or employers; the decision about termination; the risk of the procedure itself; the psychological impact of a result; and concerns about selecting for non-medical traits.

Gene therapy inserts a normal allele into affected cells. **Somatic** therapy treats body cells, so the change is **not inherited** and treatment must be repeated as cells are replaced. **Germ line** therapy would affect all cells of the resulting individual and be **heritable**, which is why it is prohibited in most countries. For CF, the normal CFTR allele is wrapped in **liposomes** or carried by a modified virus and delivered to the lung epithelium by an aerosol.

4 • The calculations that actually come up

Every one of the 22 papers contained calculations, averaging just under four per paper. Drawn from the actual papers, these are the types that recur:

Type	What to do	Seen in the papers as
Percentage and percentage change	Change ÷ original × 100. The commonest error is dividing by the final value.	Percentage decrease in new cases; percentage of genetic codes coding for amino acids
Ratios	Divide both numbers by the smaller; express in the order asked ; simplify.	Ratio of deaths caused by cancer to those not; ratio of haemophilia A to B cases
Rate	Change ÷ time, with units. For a curve, draw a tangent and use its gradient.	Rate of reaction at a given substrate concentration; rate of diffusion; rate of decrease in clotting time
Cardiac cycle timing	Read durations off a pressure–time graph. Heart rate = 60 ÷ cycle length in seconds.	Duration of one cardiac cycle; proportion of the cycle spent in ventricular systole; volume of blood passing through the heart in a given time
Surface area to volume ratio	Compute both, then express as SA : V or divide. Watch for cylinders and spheres — formulae are given.	SA:V ratio of a hydra; volume of a liposome from a given formula
BMI and waist-to-hip ratio	BMI = mass (kg) ÷ height (m) ² . Convert cm to m first .	WHR of a named individual; waist size from a given ratio
Water potential	Ψ = Ψ _s + Ψ _p . Watch the negative signs .	Water potential of a tissue with a given turgor pressure
Magnification and actual size	Actual size = image size ÷ magnification. Convert units before dividing.	Dimensions from micrographs and scale bars
Risk and probability	Express as a fraction, percentage or "1 in n" — as the question asks .	Risk of developing CVD within a period; probability of an offspring genotype

FOUR RULES THAT RECOVER MOST LOST CALCULATION MARKS

1. Show every step. Method marks survive an arithmetic slip; a bare wrong answer scores nothing.
2. Give the **units**, and the number of **significant figures or decimal places** asked for.
3. Convert units **before** calculating, not after.
4. Sanity-check the magnitude. A heart rate of 4000 bpm or a BMI of 0.02 means a unit error.

5 • Mark-scheme phrase bank

These phrases were counted across the 22 mark schemes and are the ones that recur most. The left column is what scores; the right is the common near-miss that does not.

Context	Write this	Not this
Enzyme action	"The active site is complementary to the substrate, forming an enzyme-substrate complex "	"The substrate fits into the enzyme"
Denaturation	" Hydrogen and ionic bonds holding the tertiary structure break, so the active site changes shape and is no longer complementary "	"The enzyme is destroyed / killed / melts"
Any exchange surface	" Large surface area, short diffusion distance, and a steep concentration gradient maintained "	"It is thin and big"
Osmosis	"Water moves from a higher water potential to a lower water potential through a partially permeable membrane "	"Water moves from high to low concentration"
Active transport	" Against the concentration gradient , using a carrier protein and ATP "	"It uses energy to move things"
Base pairing	" Complementary base pairing : A with T by two hydrogen bonds , C with G by three "	"The bases match up"
Correlation data	"This shows a correlation , which does not prove causation ; other variables were not controlled and the sample size may be too small"	"This proves that X causes Y"
Valves	"The valve opens because the pressure behind it is greater than the pressure in front"	"The valve opens to let blood through"
Artery structure	" Elastic tissue stretches and recoils , maintaining pressure and smoothing the flow "	"The wall is thick so it is strong"
Capillary exchange	" Large total cross-sectional area slows the blood , allowing more time for exchange "	"Capillaries are small"
Polysaccharide storage	" Insoluble , so no osmotic effect ; compact ; branched , giving many ends for rapid hydrolysis "	"It stores lots of energy"
Saturated vs unsaturated	"Double bonds cause kinks , so the chains pack less closely "	"Unsaturated fats are healthier"
Mutation effect	"The primary structure changes, so the tertiary structure and therefore the active site / function changes"	"The protein stops working"
Degenerate code	"The code is degenerate , so the new codon may code for the same amino acid and the protein is unchanged"	"Nothing happens"
Describing a graph	Quote figures with units from the axes, and name the trend — "rises from 2.1 to 6.8 between 10 and 40 minutes, then plateaus"	"It goes up then stays the same"

6 • The mistakes that cost the grade

1. Writing a **description** when the command word was **Explain**. Nearly ten "explain" marks per paper depend on this.
2. Leaving **Suggest** questions blank because the content was unfamiliar. They are designed to be unfamiliar.
3. Answering a **Compare** question about only one of the two things.
4. Not quoting **figures with units** when describing data from a graph or table.
5. Saying data **proves** a causal link. It shows a correlation.
5. Saying an enzyme is **killed** or **destroyed** rather than **denatured**.
7. Claiming denaturation breaks **peptide bonds**. It breaks the hydrogen and ionic bonds maintaining tertiary structure.
8. Defining osmosis by **water concentration** instead of **water potential**.
9. Confusing **active transport** with **facilitated diffusion** — the distinguishing features are ATP and movement against the gradient.
10. Getting **LDL and HDL the wrong way round**. LDL carries cholesterol **to** the tissues and raises risk.
1. Describing the clotting cascade out of order, or omitting **calcium ions**.
2. Forgetting that **fibrinogen is soluble** and **fibrin is insoluble** — that contrast is often the mark.
3. Confusing **amylose** (unbranched, 1,4 only) with **amylopectin** (branched, 1,4 and 1,6).
4. Saying unsaturated fatty acids have "more" double bonds without saying this causes **kinks** that prevent close packing.
5. In a genetic diagram, omitting the **key defining the symbols** or the **gametes**.
5. Giving a ratio in the wrong order, or not simplifying it.
7. Dividing by the **final** value instead of the **original** in a percentage change.
3. Not converting **cm to m** before calculating BMI.
9. Describing CF symptoms without the **chloride ion → sodium → water by osmosis** chain.
10. Writing "**the DNA unzips**" without naming **helicase** and the **hydrogen bonds**.
1. Saying mRNA is made from the "DNA strand" without specifying the **template or antisense** strand.
2. Confusing **codon** (on mRNA) with **anticodon** (on tRNA).
3. Over-writing on Q1–Q4 and running short of time for the heavily weighted Q7 and Q8.
4. In an evaluation question, giving both sides but **never stating a judgement**.
5. Leaving the extended-response question until last and writing it in a rush. Its marks are awarded for **logical structure** as well as content.

7 • Final checklist

Work through this two days before the exam. Anything you hesitate on is where the remaining revision should go.

Can you, from memory...	Can you, from memory...
• Give five properties of water with their biological importance? • Distinguish amylose, amylopectin and glycogen by bond type and branching? • Give the four-point answer on why polysaccharides suit storage? • Describe triglyceride formation, naming the bond and the water released? • Explain saturated versus unsaturated in terms of kinks and packing? • Describe all three stages of the cardiac cycle with the valve movements? • Explain why a valve opens, in terms of pressure? • Match artery, capillary and vein structure to function? • Explain tissue fluid formation and return? • Give the five-step atherosclerosis sequence including positive feedback? • Write the clotting cascade in order, with calcium ions? • Give five reasons a correlation does not establish causation? • Explain LDL and HDL and their opposite effects on risk? • Name four drug classes with mechanism and side effects?	• State Fick's law and map three alveolar adaptations onto it? • Describe the fluid mosaic model including cholesterol and glycoproteins? • Distinguish all five transport processes by energy and direction? • Define osmosis using water potential? • Give the full CFTR chain from gene to thick mucus? • Explain CF effects on all three body systems? • Give the four levels of protein structure with their bonds? • Describe collagen's structure and why glycine matters? • Explain temperature and pH effects on enzymes using tertiary structure? • Distinguish competitive from non-competitive inhibition? • Describe semi-conservative replication naming both enzymes? • Explain the Meselson–Stahl results and what each rules out? • Describe transcription and translation with correct terminology? • Explain why a substitution often has no effect? • Draw a full genetic diagram with key, gametes and ratio? • Compare amniocentesis, CVS and PGD, and list four ethical issues?

THE SHORT VERSION

Half the marks on this paper are explanation marks, and the mark schemes reward a small, fixed set of precise phrases. Learn those phrases, always give a reason where one is asked for, quote figures with units from every graph, show every line of every calculation, and protect your time for Q7 and Q8 where a quarter of the paper's marks sit. That is most of the distance between a good grade and a top one.